Level 1

Thinking Process (**Steps**):

1. Correct answers of each question_id are not given, so it can be found from context. This is relevant because a particular kind of question has a particular kind of answer— if most responses are A, B, C, or D these are the kinds of responses that should be valid. But something like 'borrow=>(EMPTY), lend=>(EMPTY),....' is an anomaly because that kind of response shouldn't belong as an answer for that question, no matter the score.

2. Since a correct score<-> response relation is not given for all questions, it must be found. I plan to go on a question_id, calculating the sum of the scores for a particular response, say "A", and counting how many times that response appeared.

3. Then, by calculating total_score / count I see which answer should be correct and how much mark should be given for a correct answer.

4. This information is then hardcoded and must be compared for each response to see whether it has been given the correct score. The anomalies are then separated out.

Steps 1, 2 and 3:
First I see what are the unique scores given to responses of different question_id. Assuming there is no negative marking, the highest scoring for each question is as follows:
1,2,2,2 in the given order. The -1 is by an anomaly ofcourse.

```
question_id
12563                    [0.0, 1.0, 2.0, -1.0]
12595            [2.0, 0.666667, 0.0, 1.33333]
17404            [1.33333, 0.666667, 2.0, 0.0]
17495      [1.0, 2.0, 1.5, 0.0, 0.5, 0.666667]
```

Screenshot 1

Next, I did filtering by different question_ids and then grouping them by response followed by summing the corresponding scores (total_score) and counts (total_count) and taking their ratio (total_score / total_count) to see roughly the score for each response.

Question_id = 12563

	question_id	response	total_score	total_count	ratio
7	12563	D	0.0	24282	0.000000
6	12563	C	2.0	20960	0.000095
2	12563	A	16913.0	16914	0.999941
4	12563	B	6.0	16496	0.000364
0	12563	1=>(EMPTY), 1=>3, 2=>(EMPTY), 2=>2, 3=>(EMPTY), 3=>1, 4=>(EMPTY), 4=>4	1.0	1	1.000000

Screenshot 2

Clearly, the most correct answer is ‘A’ with a score of 1 as seen in the ratio of response A in Screenshot 2 and from Screenshot 1. All other responses should have been marked 0.

For question_id = 12595

	question_id	response	total_score	total_count	ratio
18	12595	borrow=>[Receiving money with an agreement to repay it in the future, usually with interest charged.], lend=>[Giving money to a person or a business with the expectation that it will be repaid.], trade=>[The transfer of goods and services, often in exchange for money from one individual or organization to another.]	49330.000001	24804	1.988792
13	12595	borrow=>[Giving money to a person or a business with the expectation that it will be repaid.], lend=>[Receiving money with an agreement to repay it in the future, usually with interest charged.], trade=>[The transfer of goods and services, often in exchange for money from one individual or organization to another.]	4703.335685	7088	0.663563
20	12595	borrow=>[Receiving money with an agreement to repay it in the future, usually with interest charged.], lend=>[The transfer of goods and services, often in exchange for money from one individual or organization to another.], trade=>[Giving money to a person or a business with the expectation that it will be repaid.]	1447.334057	2192	0.660280
37	12595	echanges=>[Transférer des biens et des services, souvent en échange d'argent, d'un individu à un autre ou d'une organisation à une autre.], emprunt=>[Recevoir de l'argent en ayant pour accord de le rembourser plus tard, généralement avec des intérêts.], pret=>[Donner de l'argent à une personne ou à une entreprise en s'attendant à ce qu'il soit remboursé.]	2564.000000	1293	1.982985
22	12595	borrow=>[The transfer of goods and services, often in exchange for money from one individual or organization to another.], lend=>[Giving money to a person or a business with the expectation that it will be repaid.], trade=>[Receiving money with an agreement to repay it in the future, usually with interest charged.]	670.667002	1016	0.660105

Screenshot 3

Here each response has subparts and is given in 2 languages. Clearly the correct answers are:

In <u>English</u>:
borrow=>[Receiving money with an agreement to repay it in the future, usually with interest charged.],
lend=>[Giving money to a person or a business with the expectation that it will be repaid.],
trade=>[The transfer of goods and services, often in exchange for money from one individual or organization to another.]

In <u>French:</u>
echanges=>[Transférer des biens et des services, souvent en échange d'argent, d'un individu à un autre ou d'une organisation à une autre.],
emprunt=>[Recevoir de l'argent en ayant pour accord de le rembourser plus tard, généralement avec des intérêts.],
pret=>[Donner de l'argent à une personne ou à une entreprise en s'attendant à ce qu'il soit remboursé.]

This is because not only the definitions make sense in each language, the ratio value of each of these as in Screenshot 3 is close to 2— the highest grade possible for this question as shown in Screenshot 1.

Thus, for each subpart of this question (i.e. borrow answer, lend answer, etc.) **we must award 2 * ⅓ = ⅔** because responses could be given **in only one language** as seen in Screenshot 3 after grouping.

Similarly, for question_id = 17404,
correct subpart answers = ['1=>(EMPTY)', ' 1=>2', ' 2=>(EMPTY)', 2=>5',' 3=>3', ' 3=>1', ' 4=>4', ' 5=>(EMPTY)',' 6=>6'] each carrying 2/ 9 of the grade (assumed).

	question_id	response	total_score	total_count	ratio
33	17404	1=>(EMPTY), 1=>2, 2=>(EMPTY), 2=>5, 3=>3, 3=>1, 4=>4, 5=>(EMPTY), 6=>6	25101.333330	12551	1.999947
8	17404	1=>(EMPTY), 1=>1, 2=>(EMPTY), 2=>3, 3=>(EMPTY), 3=>2, 4=>4, 5=>5, 6=>6	0.000000	4059	0.000000
10	17404	1=>(EMPTY), 1=>1, 2=>(EMPTY), 2=>5, 3=>3, 3=>2, 4=>4, 5=>(EMPTY), 6=>6	2468.001234	3702	0.666667
31	17404	1=>(EMPTY), 1=>2, 2=>(EMPTY), 2=>3, 3=>(EMPTY), 3=>1, 4=>4, 5=>5, 6=>6	4481.322130	3361	1.333330
9	17404	1=>(EMPTY), 1=>1, 2=>(EMPTY), 2=>4, 3=>3, 3=>2, 4=>(EMPTY), 5=>5, 6=>6	0.000000	680	0.000000

Screenshot 4

Finally, for question_id = 17495,
correct subpart answers = ['1=>1', '1=>5', '2=>2', '2=>1', '3=>3', '3=>1', '4=>4', '4=>6', '5=>5', '6=>6'], each carrying **2/10** of the grade (assumed).

	question_id	response	total_score	total_count	ratio
245	17495	1=>1, 1=>5, 2=>2, 2=>1, 3=>3, 3=>3, 4=>4, 4=>6, 5=>5, 6=>6	22303.5	14869	1.500000
234	17495	1=>1, 1=>5, 2=>2, 2=>1, 3=>3, 3=>1, 4=>4, 4=>6, 5=>5, 6=>6	23412.0	11707	1.999829
239	17495	1=>1, 1=>5, 2=>2, 2=>1, 3=>3, 3=>2, 4=>4, 4=>6, 5=>5, 6=>6	13354.5	8903	1.500000
275	17495	1=>1, 1=>5, 2=>2, 2=>3, 3=>3, 3=>1, 4=>4, 4=>6, 5=>5, 6=>6	4830.0	3220	1.500000
282	17495	1=>1, 1=>5, 2=>2, 2=>3, 3=>3, 3=>3, 4=>4, 4=>6, 5=>5, 6=>6	2477.0	2477	1.000000

Screenshot 5

Step 4:
All of this info is hardcorded in q_dic (see below). Then, for each question_id I simply calculate the true score with the true_scorer function and collect all the anomalies together in an anomalies dataframe. The true scorers gives corresponding subpart scores by seeing whether the response's subpart is in the list of true answer lists. The complete code is given below and a csv file is attached storing the anomalies dataframe. A screenshot is shown below:

Anomalies dataframe

	taqr_id	question_id	response	Original score	New score
0	107730747	12563	B	2.0	0.0
1	26972882	12563	C	1.0	0.0
2	26987614	12563	1=>(EMPTY), 1=>3, 2=>(EMPTY), 2=>2, 3=>(EMPTY), 3=>1, 4=>(EMPTY), 4=>4	1.0	0.0
3	27044947	12563	1=>(EMPTY), 2=>(EMPTY), A=>1, B=>(EMPTY), C=>(EMPTY), D=>(EMPTY), E=>(EMPTY), F=>(EMPTY), H=>2, I=>(EMPTY), J=>(EMPTY)	2.0	0.0
4	27137859	12563	C	1.0	0.0
...	...	...	...	...	...
50238	31966014	17495	1=>1, 1=>5, 2=>2, 2=>3, 3=>3, 3=>3, 4=>4, 4=>6, 5=>5, 6=>6	1.0	1.8
50239	50148364	17495	1=>1, 1=>5, 2=>2, 2=>1, 3=>3, 3=>3, 4=>4, 4=>6, 5=>5, 6=>6	1.5	2.0
50240	52530670	17495	1=>1, 1=>5, 2=>2, 2=>3, 3=>3, 3=>3, 4=>4, 4=>6, 5=>5, 6=>6	1.0	1.8
50241	52748837	17495	1=>1, 1=>6, 2=>2, 2=>3, 3=>3, 3=>2, 4=>4, 4=>4, 5=>5, 6=>6	0.0	1.4
50242	56749770	17495	1=>1, 1=>5, 2=>2, 2=>1, 3=>3, 3=>3, 4=>4, 4=>6, 5=>5, 6=>6	1.5	2.0

50243 rows × 5 columns

Screenshot 6

Final Code

```
'''Combined Solution for all Question_Ids'''
''' Form of entries in q_dic
   question_id : [
               score per subpart,
               [..list of correct subparts...]
               ]
 '''
q_dic = {
        12563:[1, # max score is 1, so each subpart is 1 because only one option to
choose
               ['A']],
        12595:[2/3, # max score is 2, so each subpart is 2/3 because each response
is only in one language (only 3 subparts)
                    ['borrow=>[Receiving money with an agreement to repay it in the
future, usually with interest charged.',
                    'lend=>[Giving money to a person or a business with the
expectation that it will be repaid.',
                    'trade=>[The transfer of goods and services, often in exchange
for money from one individual or organization to another.]',
                    "echanges=>[Transférer des biens et des services, souvent en
échange d'argent, d'un individu à un autre ou d'une organisation à une autre.",
                    "emprunt=>[Recevoir de l'argent en ayant pour accord de le
rembourser plus tard, généralement avec des intérêts.",
                    "pret=>[Donner de l'argent à une personne ou à une entreprise
       en s'attendant à ce qu'il soit remboursé.]"]],
        17404:[2/9,['1=>(EMPTY)', # max score is 2,
                    ' 1=>2',       # so each subpart is 2/9
                    ' 2=>(EMPTY)',
                    ' 2=>5',
                    ' 3=>3',
                    ' 3=>1',
                    ' 4=>4',
                    ' 5=>(EMPTY)',
                    ' 6=>6']],
        17495:[2/10,['1=>1', # max score is 2,
                    ' 1=>5',  # so each subpart is 2/10
                    ' 2=>2',
                    ' 2=>1',
                    ' 3=>3',
                    ' 3=>1',
                    ' 4=>4',
                    ' 4=>6',
                    ' 5=>5',
                    ' 6=>6']]
```

```
        }

def true_scorer(response,grade,true_ans,splt):
    '''
        returns the true score for this response
        grade: score for each subpart of response
        true_ans: the list of true subpart answers as in q_dic
    '''
    subparts = response.split(splt)
    score = 0
    for part in subparts:
        if part in true_ans:
            score +=grade
    return score # the grade which should be given for this response.

rows = []
for question_id in q_dic.keys():

    # responses to question_id = 12595 have commas inside each subpart
    # so they must be separated using other splitting string splt i.e. "], "
    # Others are separated by commas: ","
    if question_id == 12595:
        splt = "], "
    else:
        splt = ","

    res_df = df[df["question_id"] == question_id] # filtering responses of one
particular question_id
    for row in res_df.itertuples(index=False):

        true_score = np.round(true_scorer(str(row.response),
q_dic[question_id][0],q_dic[question_id][1], splt),3) # rounding to two decimal
places

        if np.round(row.score,3) != true_score:
            rows.append({
                "taqr_id":row.taqr_id,
                "question_id": row.question_id,
                "response": row.response,
                "Original score": row.score,
                "New score": true_score
            })
anomalies = pd.DataFrame(rows) # have to put this into a csv
```